2026 LECTURE SERIES- PGCP AI

Artificial Intelligence & Big Data

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- Inference & Resolution
- AI Planning
- Big Data
- Structured vs Unstructured
- Data Analytics
- Real-World Apps

WHAT WE COVER TODAY

01

Inference & Resolution

Propositional logic, CNF, proof by contradiction

02

AI Planning

PDDL, Neuro-Symbolic AI, how planners work

03

Big Data — The 4 V's

Volume, Velocity, Variety, Veracity + numbers

04

Structured vs Unstructured

Types, databases, data lakes, scale

05

Relevance in AI

Symbiotic relationship, LLM training data

06

Analysis vs Analytics

Retrospective vs predictive, tools, pipelines

07

Applications

Smart cities, agriculture, healthcare, finance

SECTION 01

01

Inference & Resolution

How AI uses step-by-step logic to prove and disprove facts

WHAT IS INFERENCE?

Definition

Inference is the process of deriving NEW facts from KNOWN facts using logical rules. Just like a detective who connects clues step-by-step — AI uses inference to reach conclusions without guessing.

Pattern Matching — Old AI

Memorizes patterns only. Fails on new or unseen situations. Cannot adapt.

Logical Inference — Modern AI

Uses formal rules to derive new truths. Works in brand-new situations by applying known rules. No memorization needed.

> *"If facts are known, new truths can be derived without guessing."*

A Classic Inference Chain

FACT 1

If it is raining → the ground is wet

FACT 2

It is currently raining

INFERENCE

∴ The ground MUST be wet

🏥 Healthcare: AI cross-references symptoms against thousands of medical rules to logically eliminate diseases — until only the correct diagnosis remains.

PROPOSITIONAL LOGIC — THE LANGUAGE AI THINKS IN

Before AI can do inference, it needs a formal language. That language is Propositional Logic.

¬ P

NOT (Negation)

¬Raining = 'It is NOT raining'

P ∧ Q

AND (Conjunction)

Raining ∧ Cold = 'It is raining AND cold'

P ∨ Q

OR (Disjunction)

Raining ∨ Windy = 'Raining OR windy (or both)'

P → Q

IF-THEN (Implication)

Raining → WetGround = 'If raining, then wet'

Modus Ponens — Core Inference Rule

Rule: P → Q (If P, then Q)
Fact: P is TRUE
∴ Conclusion: Q must also be TRUE

Modus Tollens — Reverse Inference

Rule: P → Q (If P, then Q)
Fact: ¬Q is TRUE (Q is FALSE)
∴ Conclusion: ¬P must also be TRUE

THE RESOLUTION PRINCIPLE — AI'S PROOF ENGINE

AI's core algorithm for proving whether a statement is TRUE or FALSE. Used in automated theorem provers, expert systems, and diagnostic AI.

01 Translate to CNF

Convert all facts & rules into Conjunctive Normal Form (CNF) — a standard format of clauses joined by AND. Each clause is a group of literals joined by OR.

Example: $(A \vee B) \wedge (\neg A \vee C)$

→ ***CNF = AND of OR-clauses***

02 Combine & Cancel

Find two clauses that contain opposite literals — one has X, the other has ¬X. Cancel them out (this is called 'resolution') to produce a new, simpler clause.

$(A \vee B) + (\neg A \vee C) \rightarrow (B \vee C)$

→ ***Cancel complementary literals***

03 Refutation (Proof by Contradiction)

To PROVE a statement S: assume ¬S (its opposite) is true, add it to the knowledge base, and keep resolving. If you derive an empty clause □, contradiction found — S must be TRUE.

□ = Statement is PROVEN TRUE ✓

→ ***Empty clause □ = Contradiction found***

🎮 Game AI (Wumpus World): AI deduces safe squares from known facts — no guessing. If no breeze felt nearby, adjacent cells are provably pit-free. Pure resolution at work.

SECTION 02

02

Advanced AI Planning

From thinking to doing — generating optimal multi-step action sequences

WHAT IS AI PLANNING?

Planning = Generating a sequence of ACTIONS to move from a Start State → Goal State without violating any environment rules.

START STATE

Robot is at Room A.
Package is at Room A.
Goal room is Room C.

ACTION SEQUENCE

1. PickUp(Package, RoomA)
2. Move(RoomA → RoomB)
3. Move(RoomB → RoomC)
4. Drop(Package, RoomC)

GOAL STATE

Robot is at Room C.
Package is at Room C.
Mission complete ✓

State Space

The set of ALL possible configurations of the world. Planning searches this space to find a path to the goal.

Operators / Actions

Each action has Preconditions (must be true to execute) and Effects (what changes after execution).

Heuristic Search

AI uses smart estimates (heuristics) to avoid exploring all states — focuses on paths more likely to reach the goal.

PDDL — THE LANGUAGE OF AI PLANNING

PDDL (Planning Domain Definition Language) is the formal language used to describe planning problems to an AI planner.

DOMAIN FILE (define the world)

```
(:action MOVE
  :parameters (?robot ?from ?to)
  :precondition
    (and (at ?robot ?from)
         (connected ?from ?to))
  :effect
    (and (at ?robot ?to)
         (not (at ?robot ?from))))
```

PROBLEM FILE (define the task)

```
(:init
  (at Robot RoomA)
  (at Package RoomA)
  (connected RoomA RoomB)
  (connected RoomB RoomC))

(:goal
  (and (at Package RoomC)
       (at Robot RoomC)))
```

⚠ Old Way Problem: Engineers had to hand-code every rule, object, and action in PDDL for every new environment. Extremely time-consuming and error-prone.

NEURO-SYMBOLIC AI — THE MODERN PLANNING APPROACH

HUMAN GOAL

"Reorganize the warehouse so fragile items are on the top shelf"

→

LLM TRANSLATOR

Converts natural language → formal PDDL code automatically

→

PLANNING ENGINE

Runs A* / STRIPS search to find optimal action sequence

→

PLAN + REPORT

Actions executed + written justification generated in English

WHY THIS COMBINATION IS POWERFUL

LLMs Alone

✓ Understand plain English
✓ Generate creative solutions
✓ Handle ambiguous instructions

✗ Hallucinate steps that don't exist
✗ Ignore physical constraints
✗ Cannot guarantee correctness

Logic Planners Alone

✓ Perfect rule compliance
✓ Guaranteed valid solution
✓ Provably correct or impossible

✗ Cannot parse English goals
✗ Require manual PDDL setup
✗ Cannot generalize automatically

Combined (Neuro-Symbolic)

✓ English input → Formal output
✓ Context-aware + rule-compliant
✓ Self-documenting plans in English

✓ Optimal step-by-step sequences
✓ Provably correct actions
✓ Works across new environments

Finance: AI reads bank statement → PDDL planner evaluates loan eligibility rules → writes step-by-step approval/rejection report with full reasoning

SECTION 03

03

Introduction to Big Data

The scale, speed, and complexity of data that changed everything

THE 4 V's OF BIG DATA

Big Data = datasets so massive, fast, and complex that traditional software (Excel, relational DBs) CANNOT process them.

VOLUME

221 ZB

total global data by 2026

The sheer scale of data collected, stored, and processed. Traditional databases fail at this scale. Requires distributed storage: HDFS, S3, Azure Data Lake.

VELOCITY

29 TB/sec

generated every second globally

The speed at which data arrives and must be processed in real-time. Requires stream-processing engines: Apache Kafka, Apache Flink, Spark Streaming.

VARIETY

5+ Types

text, video, audio, sensor, logs

Data comes in diverse formats — not just rows and columns. 90% is unstructured. Requires specialized tools for each type (NLP for text, CV for images).

VERACITY

Garbage In

= Garbage Out

Data quality, accuracy, and trustworthiness. Dirty data produces wrong AI predictions. Requires data cleaning pipelines, validation rules, anomaly detection.

BIG DATA BY THE NUMBERS — 2026

221 ZB

Total Global Data

1 ZB = 1 trillion Gigabytes

0.4 ZB

Created Every Day

402.74 million Terabytes / day

29 TB

Generated Every Second

Non-stop, 24/7 from all sources

70%

User-Generated Content

Emails, videos, social media

21 B+

IoT Devices Connected

Sensors, phones, appliances

90%

Data is Unstructured

Only AI can process this at scale

Source: Statista, IDC Global DataSphere, IoT Analytics Reports — 2026 projections

SECTION 04

04

Structured vs Unstructured Data

Types, storage systems, and why 90% of data is invisible to traditional software

STRUCTURED vs UNSTRUCTURED DATA

	📊 STRUCTURED DATA — 10% of all data		🗃 UNSTRUCTURED DATA — 90% of all data
Definition	Neat, organized data in rows & columns of a table with a fixed schema.	VS	Messy, subjective data with NO predefined format or schema.
Growth Rate	Grows steadily with business operations.	VS	Growing 3× faster than structured data globally.
Examples	Spreadsheets, SQL tables, financial ledgers, customer records.	VS	Videos, audio files, social media posts, emails, raw text docs, images.
Storage	Relational DBs, Data Warehouses (SQL Server, Oracle, PostgreSQL).	VS	Data Lakes, NoSQL DBs (MongoDB, Cassandra, Hadoop HDFS).
Who Analyzes	Business users with SQL queries, Excel, Tableau, Power BI.	VS	Requires AI, NLP, Deep Learning, and Data Scientists.

🔑 **Key Insight: While structured data powers traditional BI, the 90% unstructured data — only readable by AI — holds the greatest untapped value.**

HOW BIG DATA IS STORED — TECHNICAL ARCHITECTURE

Relational Database (RDBMS)

For Structured Data

- Tables with fixed schemas (rows + columns)
- ACID transactions: Atomic, Consistent, Isolated, Durable
- Uses SQL: SELECT, JOIN, GROUP BY, etc.
- Examples: PostgreSQL, MySQL, Oracle, SQL Server
- Scales vertically — add RAM/CPU to one server
- Best for: transactional data, financial records, CRM

Data Warehouse

For Analytical Queries on Structured Data

- Optimised for reads, not writes (OLAP vs OLTP)
- Star schema: Fact tables + Dimension tables
- Column-oriented storage for fast aggregations
- Examples: Snowflake, BigQuery, Redshift, Synapse
- Scales to petabytes for reporting & dashboards
- Best for: historical analysis, BI reports, KPIs

Data Lake

For All Data Types at Scale

- Stores raw data in original format — schema-on-read
- Handles structured, semi-structured & unstructured
- Uses distributed file systems: HDFS, Amazon S3
- Examples: Azure Data Lake, Databricks Delta Lake
- Scales horizontally — add more commodity machines
- Best for: ML training data, raw event logs, media

SECTION 05

05

Relevance of Big Data in AI

The symbiotic relationship that powers modern AI systems

BIG DATA + AI — A SYMBIOTIC RELATIONSHIP

🤖 AI NEEDS BIG DATA

- Modern LLMs (GPT, Gemini) trained on trillions of text tokens — without this data, the model has nothing to learn from.
- Computer vision models need millions of labeled images just to recognize a cat reliably.
- More diverse data = smarter, less biased AI models (mono-source data creates blind spots).
- Reinforcement learning agents need billions of simulation steps = huge data pipelines.

⟷

🗄 BIG DATA NEEDS AI

- 90% of global data is unstructured — humans cannot manually read/watch it all at scale.
- AI 'watches' 10,000 hours of security footage in minutes, flagging incidents automatically.
- NLP reads millions of product reviews → extracts sentiment without a human reading one.
- Only AI can turn massive data chaos into actionable business intelligence in real time.

Pattern Discovery

Find trends in billions of data points impossible for humans

Real-Time Decisions

Process live sensor streams to act within milliseconds

Predictive Power

Forecast future events with high accuracy using ML models

Scale & Automation

Handle workloads impossible for any human team

HOW AI MODELS ARE TRAINED ON BIG DATA

Training a large AI model is a massive data engineering exercise BEFORE any machine learning begins.

1 Data Collection

Web scraping, APIs, sensors, databases. Raw, messy, diverse.

2 Data Cleaning

Remove duplicates, fix encoding errors, filter toxic content, validate formats.

3 Data Labeling

Human annotators or auto-labelers tag data: 'This image = cat', 'This review = negative'.

4 Feature Engineering

Transform raw data into numerical representations AI can process (tokenization, embeddings).

5 Batch Training

Feed data in mini-batches through neural network. Adjust weights via backpropagation.

6 Evaluation & Iterate

Test on held-out data, measure accuracy, go back if performance is poor.

SECTION 06

06

Data Analysis vs Data Analytics

Past-looking reports vs future-predicting AI systems

DATA ANALYSIS vs DATA ANALYTICS

DATA ANALYSIS

Time Focus

RETROSPECTIVE — Looking Backward

Core Question

"What happened and why?"

Goal

Summarize history into human-readable dashboards

Tools

Excel, Power BI, Tableau, SQL, Looker

Output

Charts, dashboards, PDF reports, pivot tables

Users

Business analysts, managers, executives

V
S

DATA ANALYTICS

Time Focus

PREDICTIVE — Looking Forward

Core Question

"What will happen next and what should we do?"

Goal

Automate decisions using AI/ML models

Tools

Python, R, Spark, TensorFlow, Databricks, SageMaker

Output

Forecasts, risk scores, auto-recommendations, alerts

Users

Data Scientists, ML Engineers, strategic decision-makers

THE DATA ANALYTICS PIPELINE — TECHNICAL VIEW

How raw data becomes actionable predictions: a 5-stage technical pipeline used by every modern data team.

INGEST

Data comes in

Tools:

Apache Kafka
AWS Kinesis
Fluentd / Logstash

Stream or batch ingest from databases, APIs, IoT sensors, user events. Handle millions of events per second.

→

STORE

Raw data saved

Tools:

HDFS / S3
Delta Lake
Cassandra / MongoDB

Store raw data in data lake (cheap, scalable). Schema-on-read — decide structure later.

→

PROCESS

Clean & transform

Tools:

Apache Spark
Databricks
Apache Flink

Run distributed jobs across hundreds of servers. Clean, join, aggregate. Batch or real-time stream processing.

→

MODEL

AI learns patterns

Tools:

Scikit-learn / XGBoost
TensorFlow / PyTorch
SageMaker / Vertex AI

Train ML models on processed data. Cross-validate. Deploy best-performing model to production endpoint.

→

VISUALIZE

Insights surfaced

Tools:

Power BI / Tableau
Grafana / Kibana
Custom APIs

Expose predictions via dashboards, alerts, or REST APIs. Business teams act on AI recommendations.

SECTION 07

07

Applications of Big Data & AI

Real-world deployments across industries — with a focus on India

APPLICATIONS OF BIG DATA & AI

Smart Cities

Pune, Maharashtra — ICCC Platform

- Live CCTV → AI adjusts traffic signals dynamically
- Sensors in bins/pipes → dispatch resources automatically
- Award-winning PMC Care urban governance platform

Computer Vision · IoT · Real-time Stream Processing

Agriculture

AgriStack — India National Platform

- Unique digital ID for every Indian farmer nationwide
- GPS-mapped farm boundaries registered nationally
- Instant flood-relief disbursement — zero paperwork

GIS · Satellite Imagery · Blockchain · NLP

Law Enforcement

MARVEL AI — Maharashtra

- Analyses CCTV + digital data → predicts crime patterns
- Enables police to act proactively before crimes occur
- Identified 5,600 malnutrition cases in Sindhudurg

Facial Recognition · Predictive Analytics · NLP

Finance

Algorithmic Trading & Underwriting

- AI reviews bank statements for loan decisions in seconds
- Reads news sentiment → executes stock trades autonomously
- Compliance strategy generated when regulations change

NLP · Time-series ML · Knowledge Graphs · Planning

Healthcare

AI Diagnostics & Public Health

- Cross-references symptoms with thousands of medical rules
- Early disease outbreak detection from population data
- Bioacoustics in forests → warns villages of wildlife danger

Inference Engines · Graph AI · Audio ML · NLP

IoT & Infrastructure

21 Billion+ Connected Devices

- Real-time monitoring of power grids and water systems
- Predictive maintenance: fix issues before breakdowns occur
- Smart home devices learning behavioural patterns over time

Stream Processing · Anomaly Detection · Edge AI

KEY TAKEAWAYS

01 Inference & Resolution

AI derives new facts from known rules using propositional logic. Resolution proves truth by contradiction. No memorisation — pure deduction.

02 AI Planning

Neuro-Symbolic AI combines LLMs (understand English) with logic planners (strict rules) to generate correct multi-step action sequences.

03 Big Data Scale

221 ZB globally by 2026. 29 TB every second. Traditional databases fail at this scale — distributed systems (Hadoop, Spark, Kafka) take over.

04 Structured vs Unstructured

Only 10% of data is structured (DBs). The 90% — video, text, audio — holds the most value but REQUIRES AI to unlock it.

05 Analysis vs Analytics

Analysis = Rearview mirror (Excel, dashboards, what happened). Analytics = GPS (Python, ML, what will happen and what to do next).

06 Real-World Impact

From dynamic traffic lights in Pune to instant crop-loss relief for farmers — AI + Big Data are solving massive societal challenges RIGHT NOW.

Thank You

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